ISL PA Team A

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Is education the way ?

Dataset 2

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Problem Statement

Is education the way to earn more income or does the data suggest something else, what factors can change this?

CLAUSE 1

Is education THE way?

A comparison. So we rank every variable in the file.

HYPOTHESIS 1

Education is positively associated with income, and more strongly than other variables.

CLAUSE 2

Or something else?

Is another factor doing more of the work?

HYPOTHESIS 2

A substantial part of that effect runs through which occupation you end up in.

CLAUSE 3

What can change this?

For whom does education pay more, or less?

HYPOTHESIS 3

A gap between men and women persists within education levels.

The Data

48,842
employed adults, one per row

14 + 1
employed adults, one per row

23.9 %
Lbase rate — clear $50K

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01 48,842 rows from the 1994 US Census. Each row is one employed adult: 14 characteristics plus whether they earn more or less than $50K. 11,687 earn more — a base rate of 23.9% that every group figure is compared against.

02 fnlwgt is a sampling weight. It describes the survey's design, not the person, so it cannot predict income. Capital-gain and capital-loss are leakage: the income label was derived from Adjusted Gross Income, which already includes capital gains. Evidence: 244 rows sit at exactly 99999 (the column's maximum) and all 244 are labelled >$50K. A pile-up at the maximum with a 100% outcome rate is a censoring code, not a real dollar amount.

03 Dropped from the models: the data cannot tell whether marriage leads to higher income or higher income leads to marriage. Kept for one descriptive comparison, because it turns out to explain most of the gender gap.

04 The dataset contains only employed people. If education helps a person get a job at all, that effect is invisible here — so every effect of education we report is a lower bound.

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Data Cleaning & Transformation

What?	Decision/Fix	Why?
Whitespace in raw data	Used skipinitialspace=True when loading the CSV using pandas.	To automatically strip whitespaces from the dataset upon loading.
? used as nulls	Replaced '?' strings with np.nan	To properly standardize missing values so they are not misread as actual data
Duplicate rows	Executed data.drop_duplicates()	To remove redundant identical entries from the dataset.
Unneeded columns	Dropped fnlwgt, capital-gain, capital-loss, and relationship	To remove columns that were not needed for the intended analysis.
Missing occupation data	Created a new occupation-status column (marking missing vs. present) and filled missing occupation values with 'Unknown'	To preserve the rows for analysis while explicitly flagging and managing the missing occupation data.
Missing Education data	Executed data.dropna(subset=['educational-num'])	To drop rows where important educational information was missing.
Data Binning & Grouping	Binned data into education-band, age-band (using pd.cut), and hours-band (using pd.qcut).	To group highly detailed levels and continuous values into consolidated bands for graphing.
Qualitative to Quantitative	Converted gender and income to 1 and 0	To format qualitative text data into quantitative numeric values (e.g., Male = 1, >50K = 1) for calculations.
Native country complexities	Used nested np.where conditions to replace NaNs with 'Unknown' and group the rest into 'US' or 'Non-US'	To simplify the country data into three manageable categories for subsequent tables.

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The missingness is not random

2,805 people have no recorded occupation. Are they like everyone else? We tested it.

	n	rate > $50K	Mean age	Mean hours
Occupation missing	2,805	9.45 %	40.1	31.8
Occupation present	45,985	24.83 %	38.6	41.0

Two-proportion $z = -18.53$, $p \approx 10^{-76*}$

The gap is 18.5 standard errors wide. Chance would essentially never produce it.

So dropping these rows is a selection decision, not a tidy-up. We keep the 2,805 records in every analysis that doesn't need the occupation column, and restrict only where it's genuinely needed and state what the restriction costs.

Workflow

01 DIAGNOSE

Find what's wrong before touching anything: the "?" sentinel, duplicates, target-label check, missingness test

02 CLEAN

The decision log you just saw; everydrop and recode logged with its reason

03 STRATIFY

group → rate → Wilson interval → z-test. Almost every finding comes from here

04 MODEL

One regression, at the end, only for what stratification cannot hold fixed at once

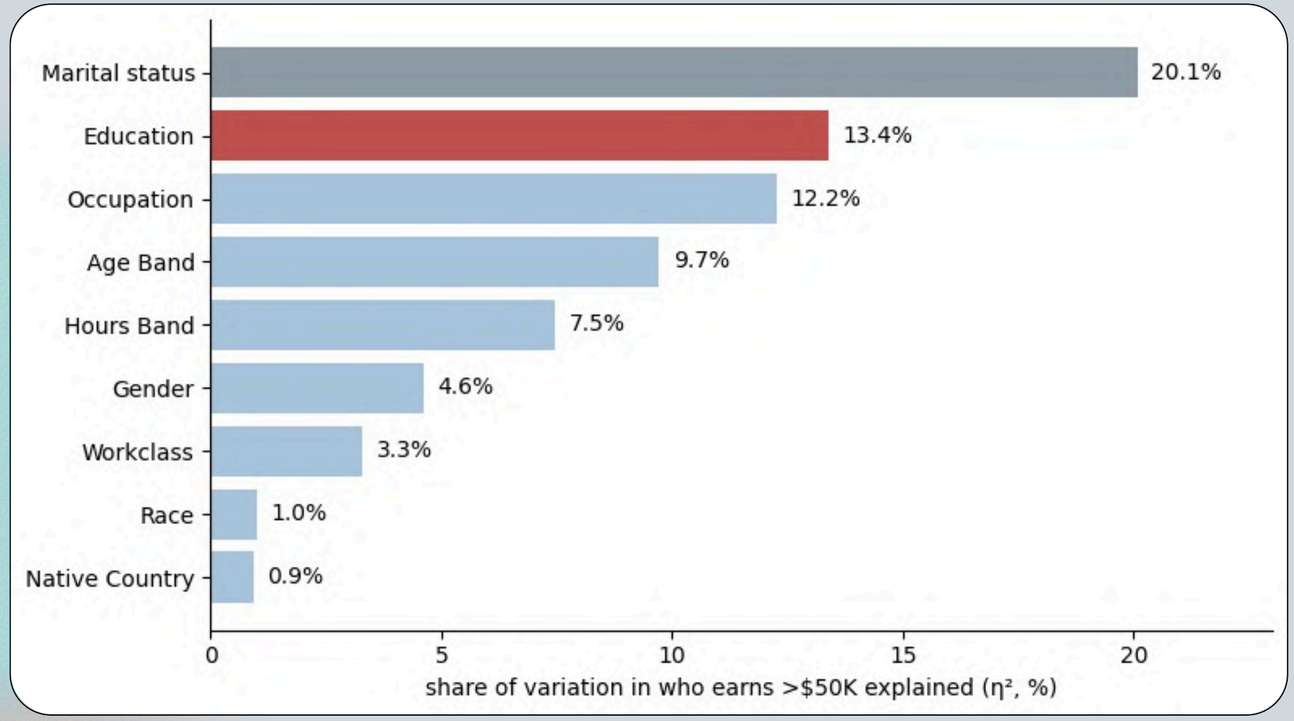
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Mathematics Used

Tool	Use Case	Why?
Using the proportion as the mean	Every rate	For a binary column, the proportion IS the mean.
Wilson Confidence Intervals	Every rate, on every graph	The standard formula gave Preschool a lower bound of −1.2%, a negative proportion.
Two-proportion z-test	Missingness, extreme categories, sex gaps	Decides whether a gap is real before we interpret it
Odds ratio	The sex gap across education levels	Doesn't depend on the base rate, so it can test whether a gap is constant
η^2 (eta-squared)	Ranking all nine variables	Weights every category by its size, no small cell can move it

Education is the strongest thing a person can acquire

Education explains 13.4%* of the variation in who clears $50K more than occupation (12.3%), age (9.7%), hours (7.5%), sex (4.6%), race (1.0%)
The share of variation in who earns >50k explained by x variable is given by eta squared
Only marital status is higher (20.1%) entangled with the extraction filter: married women who stopped working are absent from the file (17.4% of women here are married vs 62% of men). η^2 weights every category by its size; no small cell can move it. Range puts workclass 2nd on a cell of 10 people.



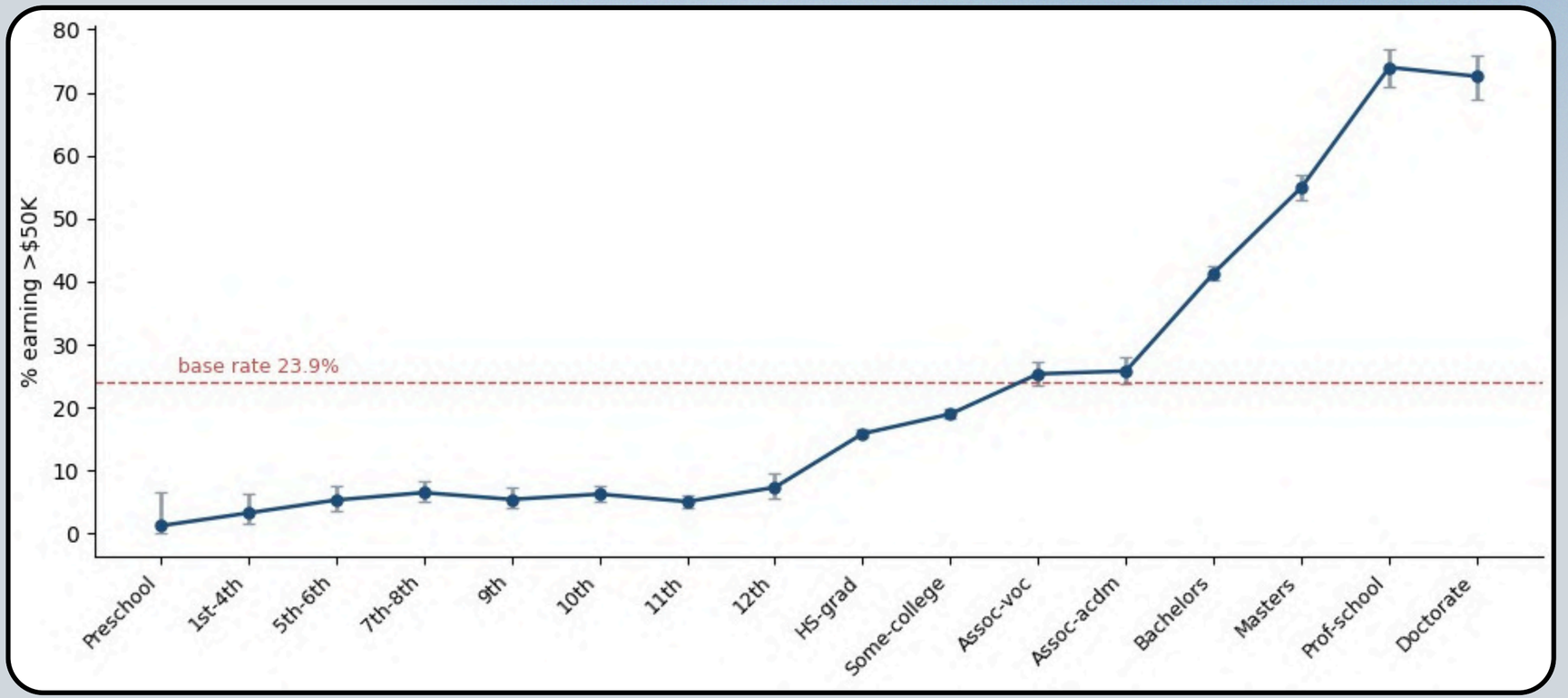
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EDUCATION GRADIENT

Level	Education	n	N >50K	Rate
1	Preschool	81	1	1.2%
2	1st-4th	245	8	3.3%
3	5th-6th	507	27	5.3%
4	7th-8th	954	62	6.5%
5	9th	756	41	5.4%
6	10th	1,389	87	6.3%
7	11th	1,812	92	5.1%
8	12th	655	48	7.3%
9	HS-grad	15,770	2,501	15.9%
10	Some-college	10,863	2,061	19.0%
11	Assoc-voc	2,060	522	25.3%
12	Assoc-acdm	1,601	413	25.8%
13	Bachelors	8,013	3,311	41.3%
14	Masters	2,656	1,459	54.9%
15	Prof-school	834	617	74.0%
16	Doctorate	594	431	72.6%

The graph showing education levels and its effect on income is flat at first, meaning that education has little effect on income from pre-school (1.2%) to 12th grade (7.3%). There is a sharp rise as we move from 12th grade to HS grad since 12th grade means you attended the final year but did not graduate whereas, HS grad means you now have a diploma and so the rate more than doubles to 15.9%.

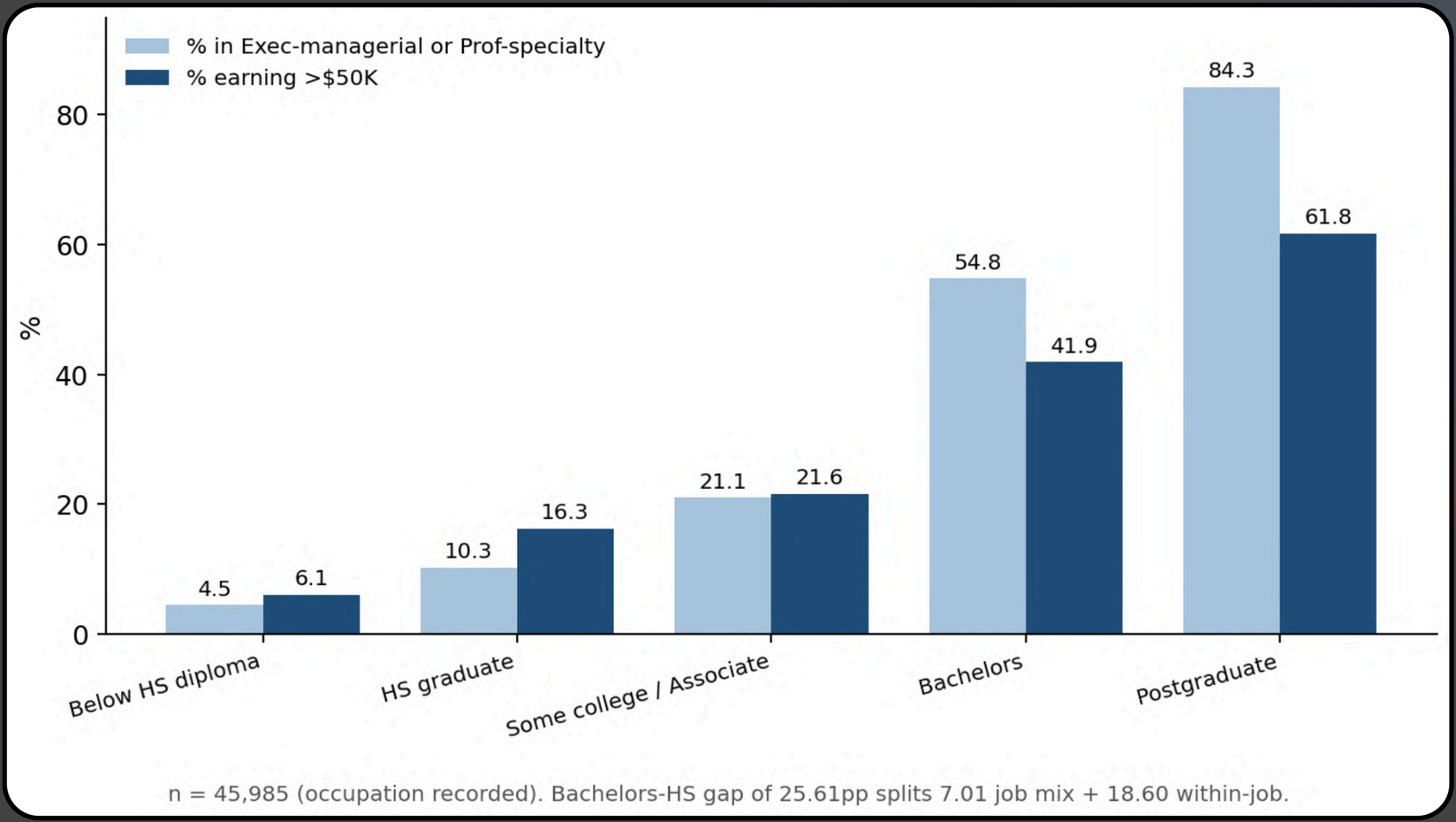
Doctorate (72.6%) , which is expected to have a higher rate than Prof- School (74.0%), sits 1.4 points below Prof-School. Since we have relatively small samples of 594 and 834 people with the two qualifications, the resulting rates will vary when resampled by some amount. Specifically, even if both groups (Doctorate and Prof-school) are considered to be identical, a gap of 1.4 pp or more happens by chance about 55% of the time. Therefore, we cannot tell if the difference is real and so we do not interpret it.



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Occupation Decomposition

A third of it is which job you end up in



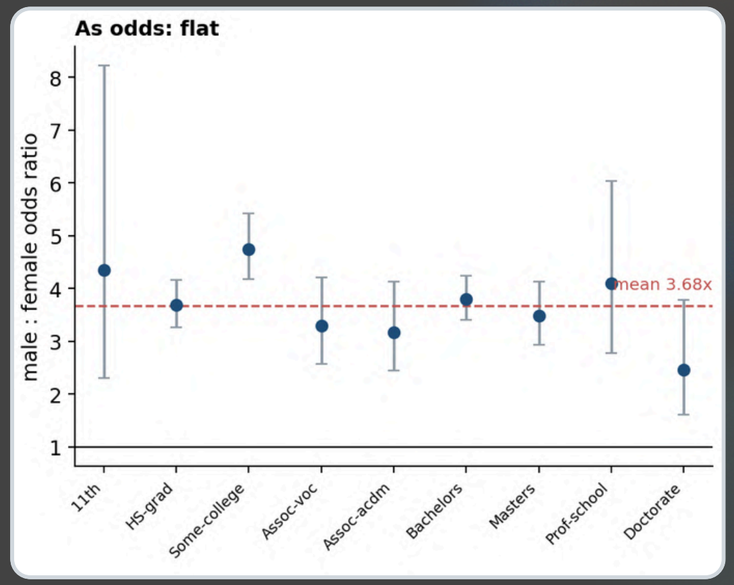
As education rises, the share in the two best-paying occupations goes **4.5% → 84.3%**, and the income rate goes **6.1% → 61.8%** together

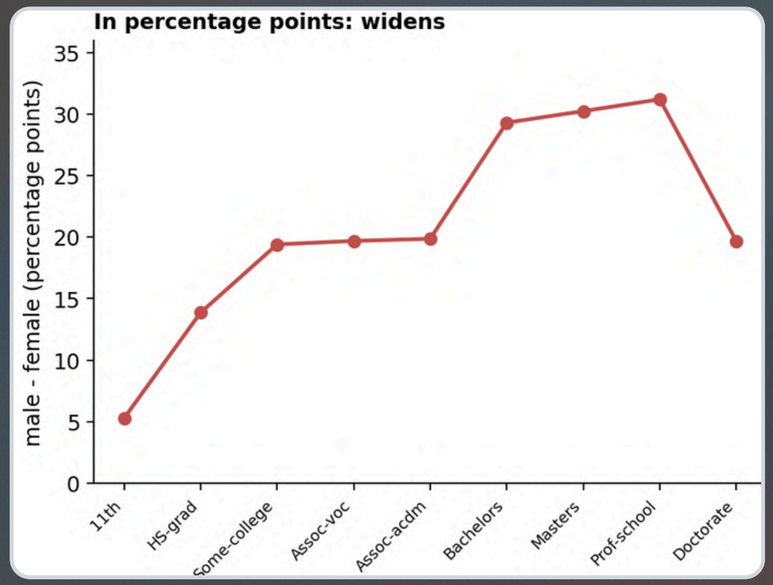
Bachelors vs HS-grads: 25.61pp gap = 7.01 (different jobs) + 18.60 (better paid in the same job).

The two parts sum exactly : an accounting identity, not a causal model.

Baseline choice gives 27%-40% via job mix so we report both.

The gap is constant, not widening





In percentage points the gap grows: 13.9 → 31.2
As an odds ratio* it's flat: ~3.7× at every level.
Education neither widens nor narrows it The gap didn't grow. The scale did.
At a 6.5% base rate, a gap involving women cannot exceed 6.5 points; pp gaps grow mechanically as base rates rise. Odds ratios shown at 9 of 16 levels, suppressed where the smaller cell has under 10 cases.

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Education slope by subgroup

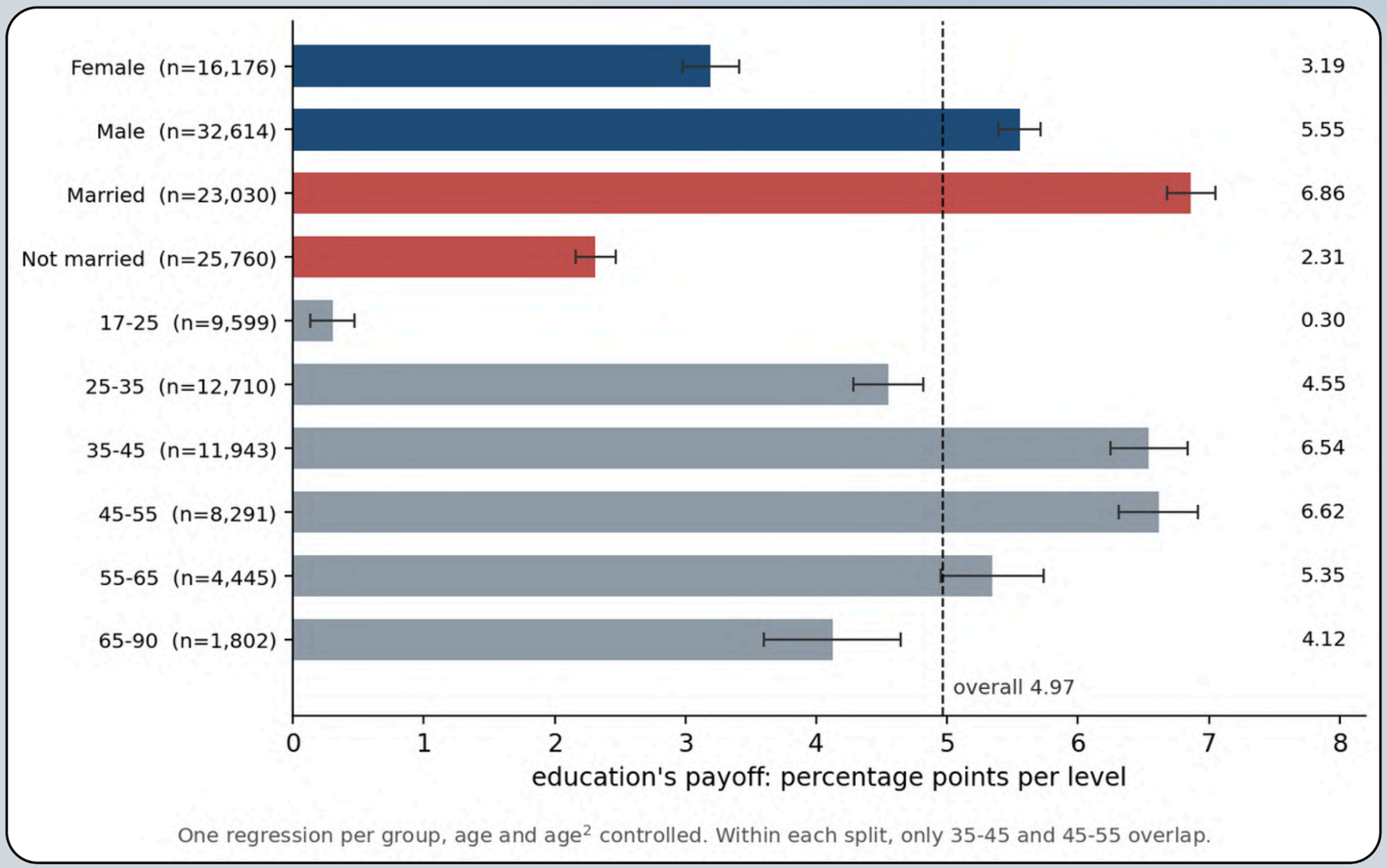
Per education level, the payoff in percentage points:

Men 5.55 vs women 3.19 · Married 6.86 vs unmarried 2.31 · Under-25 0.30 rising to 6.62 at 45–55*

The overall figure of 4.97pp per level describes almost nobody

One regression per subgroup, age and age² controlled. No single jump between levels can drive the result

The gender difference is concentrated at and below Bachelor's. Bachelors → Masters pays men 14.2pp and women 13.3pp, almost identical.



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The Answer

(a) Is education the way? Largely yes.
Education explains 13.4% of the differences in who earns over $50K, more than occupation, age, hours, sex, or race. Nothing a person can actually go and get explains more.

(b) Does the data suggest something else? Two things.
Marital status explains more (20.1%), but the dataset is missing most married women, so that number can't be trusted at face value. And a third of education's own effect isn't education directly: it's that educated people end up in better-paying occupations.

(c) What can change this? Who you are.
One more level of education is worth 5.6 points for men but 3.2 for women; 6.9 if married, 2.3 if not; 0.3 under age 25, rising to 6.6 at ages 45–55. There is no single "value of education."

⚠ This file contains only employed people. Education's effect on getting a job at all is invisible here, so every number above is an underestimate, and the data is from 1994, so it can not meaningfully serve as a basis for modern policy or conclusions about today.

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Round 2 .

Dataset 2

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Model A

Term	**Coefficient**	**In words**
Education Level	+4.97pp	per level, holding age, sex, region fixed. 95% CI 4.84–5.11
Age	+0.029	rises, then bends down — peak located at 54
Age^2	-0.000267	without it, the model believes an 80-year-old out-earns a 50-year-old
Sex= Male	16.45pp	an adjusted association — NOT causal (Table 2 fallacy)
Non-US born	-0.2pp, p = 0.76	does nothing once the rest is accounted for — we report that it did nothing

R^2 = 0.2167 — we are measuring one slope, not predicting individuals. A large, tightly estimated coefficient inside a low-R^2 model means this factor matters, and most of life is not this factor.

Deliberately absent: occupation, hours, marital status — mediators, not confounders. Conditioning on occupation opens a collider path.

Our Four Checks

Check	Result	Verdict
Restrict to the 45,985 occupation-complete rows	5.08 vs 4.97pp	restricting inflates it 2.2%; so we don't restrict
Add race to the model	4.90 vs 4.97pp	findings don't depend on excluding race
Education as 16 categories instead of a line	R^2 0.2287 vs 0.2167	the line costs ~1 point of R^2 — 4.97 is an *average slope*
Keep the 52 duplicates instead of dropping	moves ~0.01pp	irrelevant, either choice was safe

The 16-category check matters: 11th → 12th is worth +2.2pp, 12th → HS-grad +8.6pp. One coefficient asserts every level is worth the same. It isn't, and we say so.

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What the sex gap is made of

Mantel–Haenszel: the odds ratio inside each stratum, pooled by stratum size

Holding fixed	male : female odds ratio
Nothing	**3.56**
Occupation	**4.16 ↑ worse**
Education	3.89 ↑
Age	3.26
Hours	2.76
Marital Status	**1.36**
Marital status + hours	**1.03**

Selection.
62% of men here are married vs 17.4% of women. The married women in this file are heavily selected, so the 1.03 is biased toward showing no gap.

This is about clearing $50K, not hourly pay. Two people can cross that threshold on very different wages.

Hours may be a symptom, not an explanation. If women work fewer hours because of caring expectations, then "the gap is explained by hours" describes the gap rather than explaining it away.

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Limitations

THE SELECTION FILTER IS THE BIGGEST.

No one without earnings is in the file, so education's effect on getting a job at all is invisible. Every number is a lower bound. Unfixable with this data.

THE DECOMPOSITION IS DESCRIPTIVE.

It sums exactly, but the baseline choice gives 27%–40% , the index-number problem. Subgroup payoffs are effect modification, not causation.

MISSINGNESS IS OUTCOME-RELATED.

9.45% vs 24.83%. Excluding those rows from occupation analyses is a selection decision, we report both samples: 4.97 vs 5.08.

OUR INTERVALS ARE THEMSELVES A LOWER BOUND.

The census is clustered and stratified, which is exactly why the file contains fnlwgt. Clustered samples carry fewer independent observations than their row count suggests.

THE SEX COEFFICIENT IS NOT CAUSAL.

The Table 2 fallacy: our model was built to identify education; sex's confounders are a different list. men are 16.45 points more likely, and being a female causes penalty."

IT IS 1994.

Every conclusion describes the 1994 US labour market , a method and a decomposition, not current policy evidence.

family portrait











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